

Optimistic Estimate Uncovers the Potential of Nonlinear Models

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Abstract. We propose an optimistic estimate framework to evaluate the potential of nonlinear models in fitting target functions at overparameterization. In our framework, such potential is quantified by estimating the smallest possible sample size needed for a model to recover a target function, referred to as an optimistic sample size. Following the framework, we derive the optimistic sample sizes for matrix factorization models, deep models, and two-layer neural networks (NNs) with fully-connected or convolutional architectures. For each nonlinear model, we confirm via experiments that the target functions can be fitted at overparameterization as predicted by our analysis. Our results suggest a hierarchical inductive bias of nonlinear models towards simple functions with smaller optimistic sample sizes intrinsic to their architecture. The dynamical realization of the suggested inductive bias remains an open problem for the further study.

Keywords:

Nonlinear regression,
Optimistic sample size estimate,
Matrix factorization model,
Two-layer neural network,
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1 Introduction

In recent years, large nonlinear models like deep neural networks (DNNs) have achieved unprecedented success, powering amazing applications as AlphaGo [17], AlphaFold [10], and GPT [5,15]. However, for many decades, the community had severely underestimated their potential. One major factor is that traditional theoretical frameworks for generalization excessively focus on the worst-case scenarios in which overfitting easily happens in absence of regularization for overparameterized models. To circumvent the limitation of the worst-case based estimates, we propose an opposite approach of optimistic estimate

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to evaluate the best-possible fitting performance of nonlinear models at overparameterization. In this work, fitting means recovering the target function exactly.

We demonstrate the idea of the optimistic estimate using a chemical reaction analogy. By applying the law of element conservation, we derive the following optimistic estimates for chemical reactions: graphite has the potential to be converted to diamond, while gold cannot be obtained from copper. These optimistic estimates inform the potential of a chemical reaction, leading to the abandonment of alchemy and subsequently the invention of synthetic diamonds. Likewise, our optimistic estimate for nonlinear regression aims to uncover the fitting potential of nonlinear models in terms of a minimal sample size for recovering a target. Our quantitative estimate will motivate later studies comparing the empirical sample size requirement to our theoretical estimate.

To assess how well a model fits a target function, the sample size necessary for fitting serves as a natural quantity. In linear regression, the sample size is in general determined by the model size, i.e. the size of model parameters. Therefore, linear models with fewer parameters are preferred in practice. In the case of nonlinear regression, although theoretical understanding is limited, it has been empirically demonstrated that DNN models can effectively fit target functions even at heavy overparameterization [22]. To gain deeper understanding of this widely observed phenomenon, we propose an optimistic estimate framework for nonlinear models to evaluate the smallest possible sample size – which we term the optimistic sample size – necessary for fitting each target function they can express. Unlike traditional sample complexity analyses that focus on worst-case scenarios for achieving specified performance levels [16], our optimistic sample size quantifies the minimum number of training samples needed to recover a target function under ideal conditions. This approach provides new insights beyond existing work, such as Zhong *et al.*'s analysis [25] of two-layer neural network recovery based on local strong convexity assumptions, which break down in overparameterized settings.

Our results in the following show that the optimistic sample sizes of certain simple targets indeed can be much smaller than the model size for many widely considered nonlinear model families like deep models [21], matrix factorization models [2, 8, 18] and two-layer NNs. Furthermore, the optimistic sample size as a functional over the model function space exhibits rich structures, especially for the NN family.

Given a target function f^* with optimistic sample size R^* for a model f_θ , our results indicate that R^* training samples are necessary for fitting f^* . However, it remains an open problem to estimate how many samples are sufficient for fitting f^* in practice. Our experiments in matrix factorization models and two-layer tanh-NNs show that these model families can achieve near-optimism fitting performance with proper hyperparameter tuning (specifically, initialization scale and learning rate). In this context, “near-optimism” means the empirical sample size necessary for fitting f^* in experiments can approach or even match R^* . In particular, when R^* is much smaller than the model size M , the empirical sample size is generally much closer to R^* than to M . The potential mechanism for this phenomenon is discussed in Section 6.2.

Our optimistic estimates suggest quantitative answers to some long-standing problems in nonlinear regression discussed in Section 7. These problems include determining which model can fit at overparameterization, the effective size of parameters of a model, the

inductive bias of a model, the advantages of different architectures. In particular, our results suggest a hierarchical inductive bias of nonlinear models intrinsic to their model architecture. This inductive bias sets a hierarchy over the model function space, in which a function with smaller optimistic sample size is at a “higher rank”. Detailed discussions are presented in Section 6.1.

2 Optimistic sample size estimate

2.1 Optimistic initialization and model rank

We derive an optimistic estimate for a model $f_\theta = f(\cdot; \theta)$ with M parameters in the following way. In regression, we use a model f_θ to fit n data points sampled from a target function f^* , aiming to recover f^* . Without loss of generality, we consider target functions that can be expressed by model f_θ , i.e. $f^* \in \mathcal{F} := \{f(\cdot; \theta) | \theta \in \mathbb{R}^M\}$ ¹. By default, we use gradient descent to solve the regression problem

$$\min_{\theta} \frac{1}{n} \sum_{i=1}^n (f(x_i; \theta) - f^*(x_i))^2. \quad (2.1)$$

The solution is denoted as f_{θ_∞} , where θ_∞ denotes the model parameters at convergence. In the worst-case scenario, overparameterization prevents us from recovering f^* because of the non-uniqueness of the solution. However, by considering an optimistic scenario with the following best possible initialization, f^* may be recovered from less than M training samples.

Optimistic initialization: Initializing in a neighbourhood of an “optimal” point θ^* in the target set Θ_{f^*} .

Here the target set $\Theta_{f^*} := \{\theta \mid f_\theta = f^*, \theta \in \mathbb{R}^M\}$ contains all the parameter points that express the target function. Since (i) Θ_{f^*} may not be a singleton and (ii) initializing near different points in the target set may require different sample sizes for fitting (see the next section for an example), we choose to initialize near an “optimal” point in the target set such that the required sample size attains the minimum.

Remark 2.1 (Relevance of Optimistic Initialization to Global Training). Typically, a neural network is trained using a generic initialization scheme. The training trajectory that converges to a global minimum can be divided into two phases: the training phase and the convergence phase. During the convergence phase, the trajectory lies in the vicinity of a global minimum. The purpose of the training phase is to offer a suitable “initialization” for the convergence phase. In practice, while it is unrealistic to expect initialization near an “optimal” point in the target set, we observe that a highly nonlinear training phase, such as one with strong condensation in neural networks, can potentially provide an optimistic “initialization” for the convergence phase under overparameterization. In Section 6, we present preliminary numerical evidence supporting this.

¹Otherwise, a larger model should be used for fitting.

Above optimistic initialization leads to a best possible fitting scenario. It is possible because any neighborhood $N(\theta^*, \epsilon) \subset \mathbb{R}^M$ has a positive measure. It is best because any point in Θ_{f^*} recovers f^* with zero generalization error. In contrast, initializing at $\theta' \in \Theta_{f^*}$ is a best but impossible fitting scenario because Θ_{f^*} is generally a lower dimensional sub-manifold of $\mathbb{R}^M$ with zero measure [7].

We can analyze the sample size necessary for fitting in a small neighborhood $N(\theta^*, \epsilon)$ with $\theta^* \in \Theta_{f^*}$ using the linear approximation

$$f(\cdot; \theta) \approx f^{\text{lin}}(\cdot; \theta) := f^*(\cdot) + \nabla f_{\theta^*}(\cdot)^\top (\theta - \theta^*).$$

Then, the regression problem becomes

$$\min_{\theta \in N(\theta^*, \epsilon)} \frac{1}{n} \sum_{i=1}^n (f^{\text{lin}}(x_i; \theta) - f^*(x_i))^2. \quad (2.2)$$

This is a linear problem with a well-known result [28]: We need

$$n = \dim(\text{span}\{\partial_{\theta_i} f(\cdot; \theta^*)\}_{i=1}^M)$$

samples to recover f^* . This number serves as an effective number of parameters of the model at θ^* .

Inspired by the rank in differential topology, we refer to this quantity as the model rank defined as follows.

Definition 2.1 (Model Rank at a Parameter Point). *For a parameter-differentiable model f_θ and a parameter point $\theta^* \in \mathbb{R}^M$, the model rank is defined as*

$$R_{f_\theta}(\theta^*) := \dim(\text{span}\{\partial_{\theta_i} f(\cdot; \theta^*)\}_{i=1}^M), \quad (2.3)$$

where

$$\text{span}\{\phi_i(\cdot)\}_{i=1}^M = \left\{ \sum_{i=1}^M a_i \phi_i(\cdot) \mid a_1, \dots, a_M \in \mathbb{R} \right\}$$

denotes the span of basis functions, and $\dim(\cdot)$ denotes the dimension of a linear function space.

Model rank in a nonlinear model can vary across different points in the target set. In the optimistic scenario, the model rank is defined for any function $f' \in \mathcal{F}$ as the minimum rank among $\Theta_{f'}$.

Definition 2.2 (Model Rank of a Function). *For any target function $f^* \in \mathcal{F}$ with model function space*

$$\mathcal{F} := \{f(\cdot; \theta) \mid \theta \in \mathbb{R}^M\},$$

let

$$\Theta_{f^*} := \{\theta \mid f(\cdot; \theta) = f^*, \theta \in \mathbb{R}^M\}$$

denote its target parameter set. The model rank of f^* is defined as

$$R_{f_\theta}(f^*) := \min_{\theta^* \in \Theta_{f^*}} R_{f_\theta}(\theta^*). \quad (2.4)$$

Note that the notation R_{f_θ} is slightly abused for convenience, representing the model rank of either a parameter point or a function depending on its input. Under optimistic initialization, the sample size necessary for fitting f^* is $R_{f_\theta}(f^*)$. We refer to it as the optimistic sample size of f^* , which determines the minimum sample size needed to fit f^* using the model f_θ .

Remark 2.2. While optimistic initialization appears impractical since both f^* and the target set $\Theta_{f^*} := \{\theta \mid f_\theta = f^*, \theta \in \mathbb{R}^M\}$ are unknown a priori, it serves a crucial theoretical purpose: establishing a lower bound on the required sample size. The significance of this bound is twofold:

(i) It provides a fundamental limit – if function recovery fails under optimistic initialization, it is likely to also fail under any practical initialization scheme.

(ii) More surprisingly, our experimental results show that conventional small random initialization, without the need for optimistic initialization, often achieves performance remarkably close to the theoretical bound. Specifically, simple architectures such as deep models (Section 3) and matrix factorization (Section 4) models consistently reach this bound, while more complex models, like deep neural networks (Section 5), closely approach it.

2.2 Properties of model rank

The optimistic sample size defined by model rank $R_{f_\theta}(f^*)$ possesses several key properties that are fundamental to understanding its role in nonlinear regression and overparameterization:

(i) **Boundedness:** The optimistic sample size $R_{f_\theta}(f^*)$ is inherently bounded by the number of parameters M in the model

$$0 \leq R_{f_\theta}(f^*) \leq M.$$

The lower bound of zero occurs in the theoretical case where the target function f^* can be recovered without training samples, typically when the model architecture perfectly aligns with the target function with no need for parameter adjustment. The upper bound M is reached when each parameter contributes independently to the model's output, requiring the complete parameter set to recover the target function.

(ii) **Dual Dependence:** The optimistic sample size is jointly determined by the model architecture and target function. This aligns with the no free lunch theorem [20], which establishes that no single algorithm can effectively learn all possible objective functions – the complexity of an objective function must be considered in relation to the specific model attempting to learn it. A function that appears complex under one model may be simple under another. The model rank quantifies this architecture-objective function relationship, providing a mathematical framework for evaluating architectural choices. From a practical machine learning perspective, one goal is to design architectures optimally suited for specific classes of tasks. For instance, when processing images with local structure, con-

volutional neural networks (CNNs) achieve significantly lower complexity compared to fully connected networks as computed by model rank (see Section 5.1).

(iii) **Measure of Effective Parameters:** The optimistic sample size enables precise characterization of overparameterization in nonlinear regression. Traditional definitions of overparameterization (where sample size $n < M$) can be misleading. Consider fitting a linear function with $f_{\theta} = (\theta_1 + \theta_2)x$: despite having $M = 2$ parameters, this model requires only one sample for fitting. To address this limitation, we define overparameterization as the condition where $n < M_I$, where

$$M_I := \max_{\theta^* \in \mathbb{R}^M} R_{f_{\theta}}(\theta^*).$$

Here, M_I represents the maximum effective parameter count of model f_{θ} . When $R_{f_{\theta}}(f^*) = M_I$ for all $f^* \in \mathcal{F}$, the model f_{θ} cannot achieve fitting under overparameterization. For models that is linear with respect to parameters, it is easy to verify that the model rank remains constant at (M_I) , precluding their ability to recover under overparameterization. In contrast, nonlinear models – such as matrix factorization models (Section 4) and neural networks (Section 5) – can achieve recovery under overparameterization, demonstrating a fundamental advantage conferred by nonlinearity.

(iv) **Bounding the Effective Dimension.** For a symmetric matrix A of dimension k , the effective dimension is defined as

$$N_{\text{eff}}(A, \epsilon) = \sum_{i=1}^k \frac{\lambda_i}{\lambda_i + \epsilon},$$

where λ_i denotes the i -th eigenvalue and $\epsilon > 0$ is a regularization parameter [13]. Recent work by [14] has applied this concept to quantify the complexity of neural network solutions by analyzing the Hessian matrix of the empirical loss evaluated at the learned parameter point θ . Importantly, the effective dimension N_{eff} of the Hessian at a zero-loss global minimum θ is upper-bounded by our proposed model rank $R_{f_{\theta}}(\theta)$. This bound follows from the monotonic increase of effective dimension N_{eff} to the rank of Hessian as ϵ approaches 0. At a zero-loss global minimum, rank of Hessian equals the empirical model rank evaluated on the finite data samples

$$R_{f_{\theta}}^S(\theta) := \dim(\text{span}\{\partial_{\theta_i} f(\mathbf{X}; \theta)\}_{i=1}^M) = \text{rank}(\nabla_{\theta} f(\mathbf{X}; \theta) \nabla_{\theta} f(\mathbf{X}; \theta)^{\top}),$$

where

$$S = \{(x_i, y_i)\}_{i=1}^n, \quad \mathbf{X} = [x_i]_{i=1}^n, \quad f(\mathbf{X}; \theta) = (f(x_1, \theta), \dots, f(x_n, \theta)).$$

Since $R_{f_{\theta}}^S(\theta) \leq R_{f_{\theta}}(\theta)$, the effective dimension is upper bounded by the corresponding model rank.

In subsequent sections, we demonstrate these concepts through concrete examples and show experimentally that the theoretical sample complexity bounds are achievable for

simple models (such as deep models and matrix factorization) and approximately achievable in more complex architectures (such as deep neural networks).

Note that over-parameterized models can easily exhibit poor generalization, particularly with large random initialization or incorrect learning rate tuning. Therefore, to evaluate model performance under favorable conditions, we used small near-zero random initialization and carefully tuned learning rates, training with full-batch gradient descent.

3 Optimistic estimate on simple models

Initially, we apply our optimistic estimate to a basic linear model

$$f_L(\mathbf{x}; \boldsymbol{\theta}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

featuring $M = 3$ parameters. Considering any $\boldsymbol{\theta}' = [\theta'_0, \theta'_1, \theta'_2]^\top \in \mathbb{R}^3$, we observe that

$$R_{f_L}(\boldsymbol{\theta}') = \dim(\text{span}\{1, x_1, x_2\}) = 3.$$

Consequently, for any $f' \in \mathcal{F}_L := \{f_L(\cdot; \boldsymbol{\theta}) | \boldsymbol{\theta} \in \mathbb{R}^3\}$, we have $R_{f_L}(f') = M_L = 3$. Therefore, we conclude that f_L is unable to fit at overparameterization. Analogously, we can demonstrate that every linear model possesses a fixed optimistic sample size. As a result, all linear models are unable to fit at overparameterization. This finding aligns with the principles of linear regression theory.

The optimistic estimate is subject to change when we make a slight reparameterization of the linear model as

$$f_{NL}(\mathbf{x}; \boldsymbol{\theta}) = \theta_0 + \theta_1 x_1 + \theta_2 \theta_3 x_2.$$

While the function space remains the same as $\mathcal{F}_{NL} = \mathcal{F}_L$, the model becomes nonlinear in terms of its parameters. The optimistic sample size of the model is estimated as follows. Model rank as a function over the parameter space $\mathbb{R}^4$ is obtained as

$$R_{f_{NL}}(\boldsymbol{\theta}') = \dim(\text{span}\{1, x_1, \theta'_3 x_2, \theta'_2 x_2\}) = \begin{cases} 2, & \theta'_2 = \theta'_3 = 0, \\ 3, & \text{otherwise.} \end{cases} \quad (3.1)$$

Therefore, we have $M_L = 3$. By solving the minimization problem Eq. (2.4), we obtain the model rank as a function over the function space $\mathcal{F}$ as

$$R_{f_{NL}}(f^*) = \begin{cases} 2, & f^* \in \{a_0 + a_1 x_1 | a_0, a_1 \in \mathbb{R}\}, \\ 3, & f^* \in \{a_0 + a_1 x_1 + a_2 x_2 | a_2 \neq 0, a_0, a_1, a_2 \in \mathbb{R}\}. \end{cases} \quad (3.2)$$

Above estimate indicates that the nonlinear model f_{NL} can potentially fit

$$f^* \in \{a_0 + a_1 x_1 | a_0, a_1 \in \mathbb{R}\}$$

at overparameterization using two samples, but it cannot achieve the same for

$$f^* \in \{a_0 + a_1 x_1 + a_2 x_2 | a_2 \neq 0, a_0, a_1, a_2 \in \mathbb{R}\}.$$

Although the optimistic sample size derived in Section 2 provides only a lower bound and does not guarantee optimistic fitting performance, we observe a remarkable result: with well-tuned hyperparameters (specifically, the variance of Gaussian random initialization and the learning rate), many models can approach or even achieve our optimistic estimate without requiring optimistic initialization. For instance, in Fig. 3.1, we utilize gradient descent with a small initialization, wherein model parameters are randomly initialized with a small variance, to fit one, two, or three data points randomly sampled from target functions labeled on the ordinate. The color red represents a fitting (generalization) error that is close to zero. In the case of the linear model f_L , every target function requires three samples to achieve almost 0 fitting/generalization error. Conversely, for the nonlinear model f_{NL} , a mere two samples are sufficient to fit target functions $1, x_1$, and $1 + x_1$. Evidently, these simple models can achieve the optimistic fitting performance indicated by our optimistic estimate.

The above optimistic estimate can be easily extended to deep models with application in compressed sensing [21]. For example, a simple depth-2 model reads

$$f_{\theta}(x) = \sum_{j=1}^d (a_j^2 - b_j^2) x_j \quad (3.3)$$

with

$$\theta = (\{a_j\}_{j=1}^d, \{b_j\}_{j=1}^d), \quad x = [x_1, \dots, x_d]^T.$$

For this model, we have the following optimistic estimate.

Proposition 3.1. *Given the depth-2 model*

$$f_{\theta}(x) = \sum_{j=1}^d (a_j^2 - b_j^2) x_j$$

for any k -sparse target function f^* (i.e. a function with exactly k nonzero coefficients in its expression), the optimistic sample size is

$$R_{f_{\theta}}(f^*) = k. \quad (3.4)$$

The proof of this proposition is given in Appendix A (Proposition A.1).

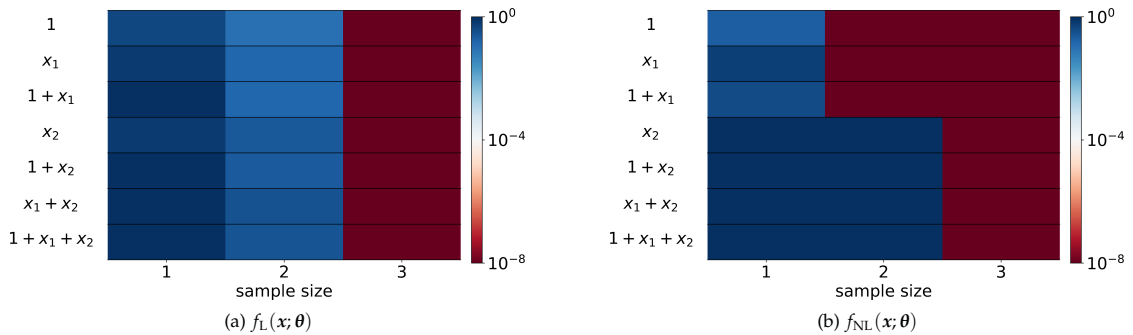


Figure 3.1: Average test error (color) vs. the number of training samples (abscissa) for different target functions (ordinate). Each test error is averaged over 100 trials with random initialization.

According to [21], this model with small initialization implicitly minimizes $\sum_{j=1}^d |\gamma_j|$, where $\gamma_j = a_j^2 - b_j^2$. By the theory of compressed sensing [6], this model can fit a k -sparse target function generally from $\mathcal{O}(k \log d)$ samples, which is much closer to $\mathcal{O}(k)$ than to $\mathcal{O}(d)$ for sparse targets with $k \ll d$.

It is observed that deep models of various forms, such as model

$$f_{\theta}(x) = \sum_{j=1}^d \left(a_j^{[1]} \cdots a_j^{[L]} \right) x_j \quad (3.5)$$

with

$$L \geq 2, \quad \theta = \left(\{a_j^{[1]}\}_{j=1}^d, \dots, \{a_j^{[L]}\}_{j=1}^d \right),$$

empirically exhibit a similar inductive bias toward sparse interpolations when trained with small initialization. However, extending the analysis of inductive bias presented in [21] to these models is challenging. In contrast, our optimistic estimate can be easily applied to these models. For model Eq. (3.5), regardless of the depth L , the optimistic sample size of a k -sparse target function remains k . This finding indicates that model Eq. (3.5) and model Eq. (3.3) belong to the same class of optimistically equivalent models, suggesting they possess the same potential for fitting at overparameterization. Generally, it is reasonable to expect models within the same optimistically equivalent class to exhibit similar inductive bias and fitting performance.

Given a generic differentiable model f_{θ} , the optimistic estimate is obtained by the following two steps: estimate $R_{f_{\theta}}(\theta)$ for $\theta \in \mathbb{R}^M$; and estimate $R_{f_{\theta}}(f)$ for $f \in \mathcal{F}$. The difficulty of optimistic estimate depends on the complexity of model architecture. In the following sections, we will present the optimistic sample sizes for several widely adopted nonlinear models, including matrix factorization models, fully-connected and convolutional NNs. Our focus lies in: (i) understanding the impact of both the model architecture and the target function on the optimistic sample size, and (ii) investigating the proximity between empirical sample sizes obtained in experiments and their corresponding optimistic sample sizes.

4 Optimistic estimate on matrix factorization models

In this section, we apply our optimistic estimate to a nonlinear matrix factorization model denoted as $f_{\theta} = AB$. This model finds application in matrix completion tasks, where the objective is to recover a target matrix M^* from n observed entries [18]

$$S = \left\{ \left((i_s, j_s), M_{i_s j_s}^* \right) \right\}_{s=1}^n.$$

To solve this problem, we employ gradient descent and minimize the following objective:

$$\min_{\theta} \frac{1}{n} \sum_{s=1}^n \left([f_{\theta}]_{i_s j_s} - M_{i_s j_s}^* \right)^2, \quad (4.1)$$

where $M^* \in \mathbb{R}^{d \times d}$, $\theta = (A, B)$, and $A, B \in \mathbb{R}^{d \times d}$.

By employing a standard optimistic estimate procedure, we have the following optimistic estimate.

Proposition 4.1. *For the matrix factorization model $f_\theta = AB$, the optimistic sample size for any $M^* \in \mathbb{R}^{d \times d}$ as*

$$R_{f_\theta}(M^*) = 2r_{M^*}d - r_{M^*}^2, \quad (4.2)$$

where $r_{M^*} = \text{rank}(M^*)$ is the matrix rank of M^* .

The proof of this proposition is given in Appendix A (Proposition A.2).

Notably, for a fixed d , the optimistic sample size increases with r_{M^*} . Particularly, for large d , the optimistic sample size for a rank- $o(d)$ target matrix is $o(d^2)$, significantly smaller than $M_I = d^2$. Consequently, the matrix factorization model can potentially fit low rank matrices at overparameterization. This finding aligns with existing researches on matrix factorization models, which suggests an inductive bias towards lower rank completions [2, 8].

We also conducted experimental tests to demonstrate the practical achievability of the optimistic sample sizes derived above. Referring to Eq. (4.2), a 4×4 target matrix M^* with a rank of 1, 2, 3, or 4 corresponds to an optimistic sample size of 7, 12, 15, or 16, respectively. Fig. 4.1 illustrates the fitting error of eight target matrices using sample sizes ranging from 1 to 16. The empirical sample size for fitting corresponds to the point at which the fitting/generalization error approaches zero for the first time. As depicted in Fig. 4.1, when employing a random initialization with a small variance, the empirical sample sizes align with the corresponding optimistic sample sizes indicated by the yellow dashes.

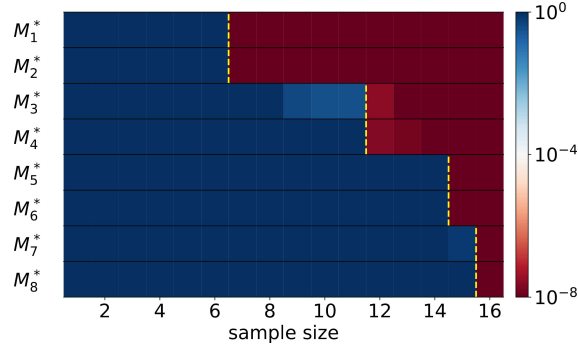


Figure 4.1: Average test error (color) vs. the number of training samples (abscissa) for different target functions (ordinate). The yellow dashed lines indicate the transitions to the corresponding optimistic sample sizes. Different target matrices are considered in different rows, where $\text{rank}(M_{2k-1}^*) = \text{rank}(M_{2k}^*) = k$ for $k = 1, 2, 3, 4$. Each test error is averaged over 50 trials with random initialization.

4.1 Role of hyperparameter tuning

Over-parameterized models are highly sensitive to their training configuration, particularly initialization scale and learning rate, with poor hyperparameter choices often leading to suboptimal generalization. As demonstrated in Fig. 4.2(a-b), large random initialization can cause the empirical sample complexity to significantly exceed our theoretical

optimistic bounds. The matrix factorization experiments in Fig. 4.2(b) specifically highlight how crucial small initialization variance is for achieving near-optimistic fitting performance.

We can directly analyze this phenomenon by examining the model rank of the learned solution, $R_{f_\theta}(\theta_\infty)$. Fig. 4.2(c) reveals that decreasing the initialization variance leads to a consistent reduction in the learned model rank, with a notable transition at rank 7 that aligns with our theoretical predictions. This observation suggests that smaller initialization naturally biases the learning process toward solutions with lower model rank. In fact, small initialization plays a crucial role in nonlinear model training by promoting model condensation – a property that effectively reduces model complexity [12]. We discuss the relationship between optimistic estimate and condensation in detail in Section 6.2.

For simpler models, such as certain deep models and matrix factorization models, initialization scale emerges as the dominant hyperparameter. Small initialization proves essential for both learning low-rank solutions and achieving optimistic sample complexity. However, neural networks present additional challenges – small initialization alone is typ-

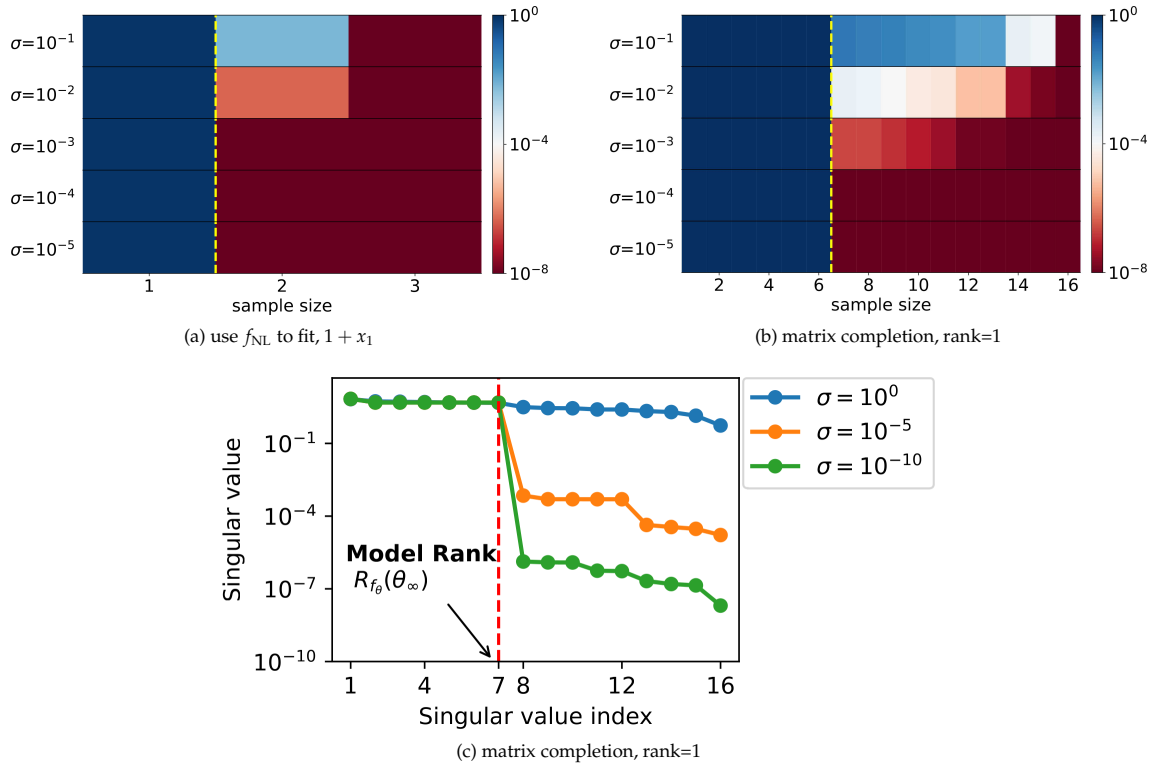


Figure 4.2: Average test error (color) vs. the number of training samples (abscissa) over different sizes of standard deviation of initialization (ordinate). (a) Performance of fitting $1 + x$ by $f_{\text{NL}}(x, \theta)$. (b) Performance of completing a rank-1 matrix by the matrix factorization model. The yellow dashed lines indicate the transitions to the optimistic sample sizes. Each test error is averaged over 50 trials with random initialization. (c) Relationship between initialization scale and learned model rank (measured by number of non-zero singular values). A significant transition occurs at rank 7 as initialization approaches zero.

ically insufficient. Achieving optimistic sample sizes requires careful tuning of multiple hyperparameters, particularly learning rates. Through joint optimization of initialization and learning rates, we have demonstrated that the required sample size can approach the theoretical optimistic bounds (see Fig. 5.1). Recent work by [24] suggests that dropout regularization can further help achieve or even match the optimistic sample size.

Alternatively, increasing the training set size beyond the optimistic sample size reduces the need for precise hyperparameter tuning. These findings align with practical observations in large nonlinear models, where both careful hyperparameter tuning and substantial training data significantly improve fitting performance in overparameterized regimes.

5 Optimistic estimate on two-layer NNs

The family of neural networks is the most significant group of nonlinear models for our optimistic estimates. However, determining their optimistic sample sizes is considerably more challenging than for the previously mentioned models. To simplify this, we focus on two-layer neural networks with activations satisfying Assumption 5.1, allowing us to easily estimate the model rank over the parameter space. Notably, the exponential activation satisfies this assumption. The commonly used tanh activation does not satisfy this assumption; however, it is linearly independent when $w_k \neq \pm w_j \neq \mathbf{0}$. Therefore, all our results remains valid for tanh activation with slight modification.

Assumption 5.1 (Linearly Independent Activation). We consider neural networks with linearly independent activations $\sigma : \mathbb{R} \rightarrow \mathbb{R}$. Specifically, for any distinct weight vectors $\{w_i\}_{i=1}^m \subset \mathbb{R}^d$ satisfying $w_k \neq w_j$ for all $1 \leq k < j \leq m$, the set of functions

$$\{\sigma(w_i^\top x), \sigma'(w_i^\top x)x_1, \dots, \sigma'(w_i^\top x)x_d\}_{i=1}^m$$

are linearly independent.

In this section, we derive the optimistic sample sizes for two-layer NNs with fully-connected or convolutional architectures. We also demonstrate via experiments that the empirical sample sizes can approach the derived optimistic sample sizes.

5.1 Optimistic sample sizes of two-layer fully-connected and convolutional NNs

We first consider a two-layer fully-connected neural network with m hidden neurons, represented by

$$f_{\theta}(x) = \sum_{i=1}^m a_i \sigma(w_i^\top x),$$

where $x \in \mathbb{R}^d$ is the input, $\theta = (a_i \in \mathbb{R}, w_i \in \mathbb{R}^d)_{i=1}^m$ denotes the network parameters, and σ is an activation function satisfying Assumption 5.1. This network has a total of $M = m(d + 1)$ parameters. For a target function f^* , we define its intrinsic width as the minimum width of an NN that can express f^* , i.e. f^* can be expressed by a width- $k(f^*)$

NN but not any narrower NN. By employing a standard optimistic estimate procedure, the optimistic sample size is a function of the intrinsic width of the target function $k(f^*)$.

Proposition 5.1. *Given the two-layer fully-connected neural network*

$$f_{\theta}(\mathbf{x}) = \sum_{i=1}^m a_i \sigma(\mathbf{w}_i^{\top} \mathbf{x})$$

with σ satisfying Assumption 5.1. For any f^* expressible by the width- m NN, denote its intrinsic width as $0 \leq k(f^*) \leq m$, the optimistic sample size of f^* is

$$R_{\text{NN}_m}(f^*) = k(f^*)(d+1). \quad (5.1)$$

The proof of this proposition is given in Appendix A (Proposition A.3).

Therefore, a function with intrinsic width $k(f^*) \ll m$ can be fitted optimistically at heavy overparameterization.

We then consider a two-layer CNN with weight-sharing represented by

$$f_{\theta}(\mathbf{x}) = \sum_{i=1}^{m_C} \sum_{j=1}^{d+1-s} a_{ij} \sigma \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{i;\alpha} \right) \quad (5.2)$$

with $\mathbf{x} = [x_1, \dots, x_d]^{\top} \in \mathbb{R}^d$. This CNN with 1-D convolution has $M = m_C(d+1)$ free parameters, where m_C is the number of convolution kernels analogous to the width of a fully-connected NN.

By employing a standard optimistic estimate procedure, similar to the fully-connected case, the optimistic sample size is a function of the intrinsic number of kernels $k_C(f^*)$ of the target function f^* .

Proposition 5.2. *Given the two-layer CNN with weight-sharing*

$$f_{\theta}(\mathbf{x}) = \sum_{i=1}^{m_C} \sum_{j=1}^{d+1-s} a_{ij} \sigma \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{i;\alpha} \right)$$

with σ satisfying Assumption 5.1. For any f^* expressible by the width- m CNN, denote its intrinsic width as $0 \leq k_C(f^*) \leq m$, the optimistic sample size of f^* is

$$R_{\text{CNN}_{m_C}}(f^*) = k_C(f^*)(d+1). \quad (5.3)$$

The proof of this proposition is given in Appendix A (Proposition A.4).

Therefore, a function with intrinsic number of kernel $k_C(f^*) \ll m_C$ can be fitted optimistically at heavy overparameterization.

5.2 Optimistic properties of two-layer NNs

Our optimistic estimate provides a means to theoretically quantifies the best possible performance of different models in fitting a target function. It is useful for model selection

and architecture design. For example, given a target function f^* , we can trace the change of its optimistic sample size with the design of model architecture to study how model architecture influences the fitting performance.

We first use this approach to study the impact of width for two-layer NNs. Given a target function with intrinsic width k , it can be expressed by all NNs with width $m \geq k$. According to Eq. (5.1), all of them yield the same optimistic sample size $k(d+1)$. Therefore, their best possible fitting performance remains the same. A similar result applies to two-layer CNNs, in which a target function with intrinsic number of kernels k can be fitted optimistically from $k(d+1)$ samples for all CNNs with $m \geq k$ kernels. Clearly, for both two-layer fully-connected and convolutional NNs, an increase in the width (or number of kernels) enhances their expressiveness without impairing the optimistic fitting performance. We call this property free expressiveness in width. This property appears to conflict with the Occam Razor principle. However, it is worth noting that the Occam Razor accounts for the worst-case scenario, while the free expressiveness property accounts for the best possible scenario. As demonstrated by our experiments, performance estimated at the best possible scenario can be close to the empirical performance especially with finely tuned hyperparameters.

We then investigate how interlayer connectivity between neurons affects the fitting performance. For instance, a two-layer fully-connected neural network connects every input neuron to a hidden neuron, while a two-layer CNN connects only a local group of input neurons to a hidden neuron. Let f^* be a target function with intrinsic number of kernels k (each kernel consist of $d+1-s$ neurons). For the convenience of discussion, we further assume the intrinsic width of f^* is $k(d+1-s)$. Then, for all CNNs with $m_C > k$ kernels,

$$R_{\text{CNN}_{m_C}}(f^*) = k(d+1).$$

For all fully-connected NNs with $m > k(d+1-s)$ neurons,

$$R_{\text{NN}_{m_C}}(f^*) = k(d+1-s)(d+1).$$

In practice, because input dimension d is often much larger than the convolution kernel size s , we have $k(d+1) \ll k(d+1-s)(d+1)$ indicating that CNNs are superior to fully-connected NNs regarding the CNN functions.

Note that both reduced connectivity and weight-sharing contribute to the smaller optimistic sample sizes of CNNs. To identify the contribution purely from the reduced connectivity between neurons, we also estimate the optimistic sample size for a no-sharing CNN.

Proposition 5.3. *Given the two-layer CNN without weight-sharing*

$$f_{\theta}(\mathbf{x}) = \sum_{i=1}^{m_C} \sum_{j=1}^{d+1-s} a_{ij} \sigma \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{ij,\alpha} \right)$$

with σ satisfying Assumption 5.1. For any f^* expressible by the width- m CNN without weight sharing, denote its intrinsic width as $0 \leq k_C(f^*) \leq m$, the optimistic sample size of f^* is

$$R_{\text{CNN}_{m_C}^{\text{NS}}}(f^*) = k_C(f^*)(d+1-s)(s+1). \quad (5.4)$$

The proof of this proposition is given in Appendix A (Proposition A.5).

Therefore, increasing the number of connections per hidden neuron from s to d multiplies the optimistic sample size by $(d + 1)/(s + 1)$. To illustrate the difference, we consider a target function expressed by a k -kernel CNN with 3×3 convolution kernel and a 28×28 input. In this scenario, the optimistic sample size is $685k$ for a CNN with 3×3 convolution kernels, $6760k$ when weight sharing is removed, $530660k$ when all neurons in adjacent layers are connected to one another². Apparently, unnecessary connections between neurons increase the NN's expressiveness at a high cost of sample efficiency. We call this property costly expressiveness in connection.

In general, we refer to the properties of models obtained from optimistic estimates as their optimistic properties. Above, we uncover two optimistic properties of two-layer NNs – free expressiveness in width and costly expressiveness in connection. These properties suggest two architecture design principles for NNs: (i) feel free to add neurons/kernels, and (ii) restrain from connecting neurons. These two principles justifies the common practice that one would rather expanding the width (or number of kernels) than adding connections to increase the expressiveness of NNs.

5.3 Numerical experiments

Fig. 5.1 presents an analysis of empirical sample sizes with fine tuning for two-layer tanh-NNs with different widths and architectures. The target function used in our experiments reads

$$f^*(x) = \sum_{s=1}^3 \tanh(0.6x_s + 0.8x_{s+1} + x_{s+2}). \quad (5.5)$$

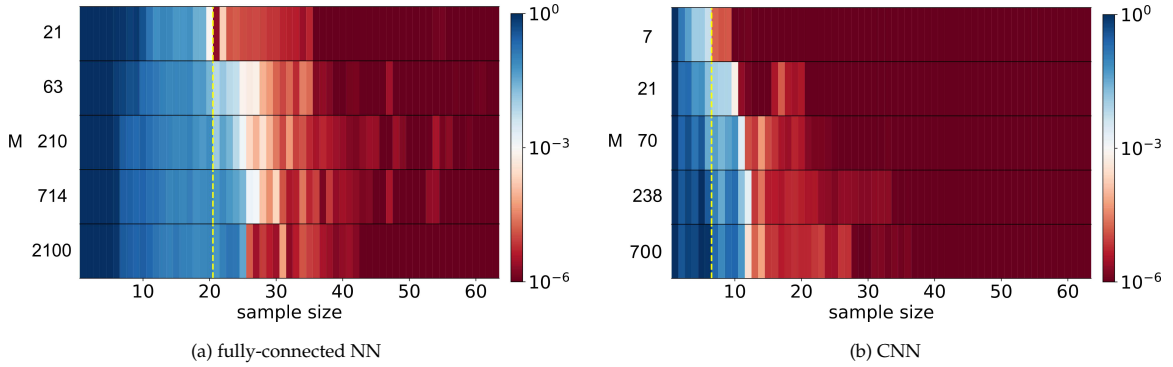


Figure 5.1: Average test error (color) for NNs with different parameter sizes (ordinate) over different training sample sizes (abscissa) in fitting the target function Eq. (5.5). Results are exhibited for (a) two-layer fully-connected tanh-NN (with bias) and (b) two-layer tanh-CNN (with bias). The yellow dashed lines indicate the transitions to the corresponding optimistic sample sizes. For each experiment, the training hyperparameters are finely tuned (see Appendices A, B for details).

²Here, we consider CNNs with 2-d convolution, whose optimistic sample size is slightly different from that for 1-d convolution (see Appendix A, Sections A.4 and A.5 for details).

We train two-layer fully-connected and convolutional tanh-NNs with various widths (parameter sizes) on randomly sampled training datasets of various sizes from 1 to 63. The learning rate for the experiments in each setup is fine-tuned from 0.05 to 0.5 for a better generalization performance. Remark that, due to the free expressiveness in width, the optimistic sample size indicated by the yellow dashed line does not increase as the size of NN expands. As shown in Fig. 5.1, for both architectures, the empirical sample sizes are close to the corresponding optimistic sample sizes over a wide range of parameter sizes. In contrast, they are often far less than the parameter sizes especially for large NNs. These results demonstrate that our optimistic estimate accounting for the best possible scenario is highly relevant to the practical performance of NNs with hyperparameter tuning. The potential mechanism underlying the near-optimism fitting performance of NNs is discussed in Section 6.2.

6 Hierarchical inductive biases and condensation

6.1 Hierarchical inductive biases

Understanding the inductive biases in overparameterized models remains a fundamental challenge in deep learning and nonlinear regression. Our analysis reveals an optimistic inductive bias that favors interpolations with lower optimistic sample sizes. Specifically, given n training samples, a nonlinear model f_θ tends to prefer interpolations $\hat{f}$ where the optimistic sample size satisfies $R_{f_\theta}(\hat{f}) \leq n$.

We validated this hypothesis through experiments on both matrix factorization models (Fig. 6.1) and neural networks (Fig. 6.2), examining how the learned solution $f_{\theta(\infty)}$ evolves with increasing sample size.

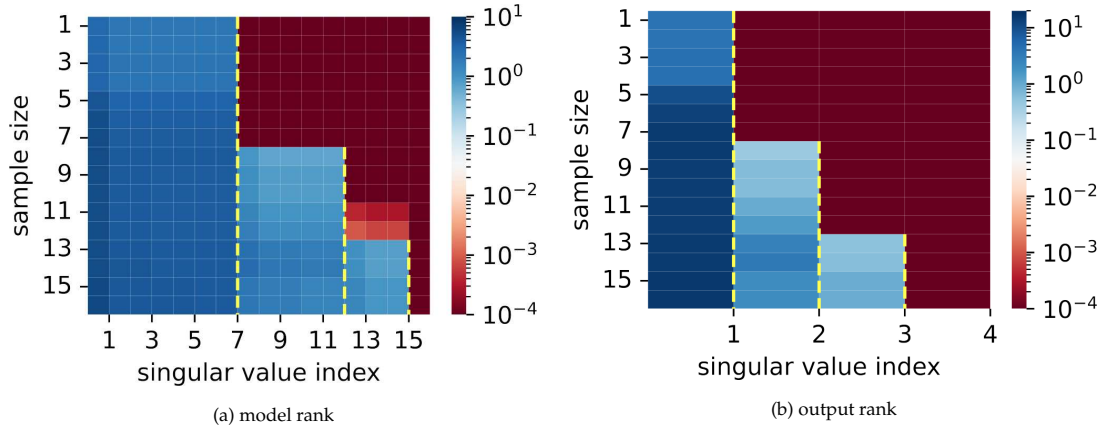


Figure 6.1: Characterization of the matrix factorization model's learning preferences as training samples increase. The target matrix is a rank-3 matrix of size 4×4 . Training was conducted under small random initialization following a fixed connected sampling pattern (see Appendix B for details). Panel (a) shows the model rank (dimensionality of the tangent space) of the converged solution $f_{\theta(\infty)}$ at near-zero training error. Panel (b) displays the matrix rank of the converged solution $f_{\theta(\infty)}$.

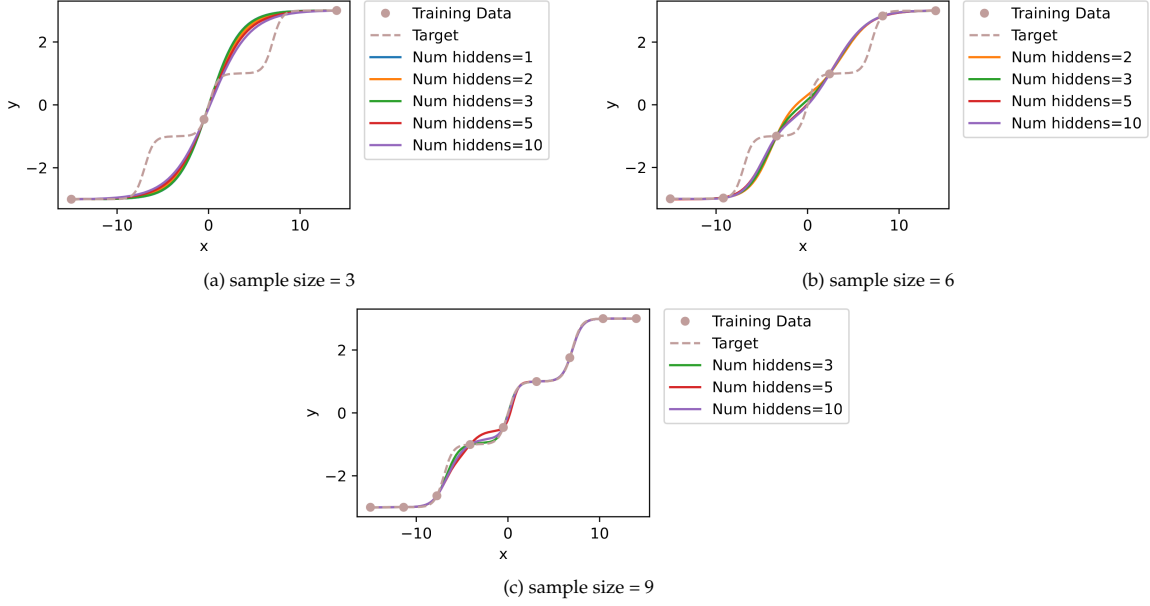


Figure 6.2: Evolution of learned solutions in a two-layer neural network model with increasing training samples. The target function consists of 3 tanh components as defined in Eq. (5.5) (shown as pink dashed lines). Training samples were uniformly drawn from the interval $[-15, 14]$ (indicated by dots in the figures). The network was trained using Adam optimizer with learning rate 10^{-3} under small random initialization (see Appendices A, B for experimental details). Panels (a-c) show the output functions learned by models of varying sizes with sample sizes of 3, 6, and 9, respectively.

For matrix factorization, we targeted a rank-3 matrix of size 4×4 , which requires $R_{f_\theta}(f^*) = 15$ samples for recovery. As shown in Fig. 6.1(a), the model exhibited distinct learning phases: with less than or equal to 7 samples, the learned model maintained rank 7; with 7-12 samples, it achieved rank 12; and with 12-16 samples, it reached rank 15. Fig. 6.1(b) demonstrates a similar hierarchical pattern in the output rank, progressing from rank 1 (less than or equal to 7 samples) to rank 2 (7-12 samples), and finally to rank 3 (12-16 samples), indicating a clear low-rank bias.

For neural networks, we examined the recovery of a target function composed of three tanh components

$$f^*(x) = \tanh(x - 7) + \tanh(x) + \tanh(x + 7), \quad (6.1)$$

using a model of the form

$$f_\theta(x) = \sum_{i=1}^m a_i \tanh(w_i^\top x).$$

Complete recovery of this function requires at least $R_{f_\theta}(f^*) = 9$ samples. Fig. 6.2 illustrates the progressive learning behavior: With 3 samples, the model approximated the function using a single tanh component; with 6 samples, it utilized two components; and with 9 samples, it achieved near-perfect recovery of the target function, demonstrating a low intrinsic width bias.

This low R_{f_θ} inductive bias manifests differently across model architectures: as a sparsity bias in deep models [21], a low-rank bias in matrix factorization, and a low intrinsic width bias in neural networks. Notably, functions with lower R_{f_θ} correspond to simpler models within their respective model classes. This creates a hierarchical inductive bias where simpler model functions are prioritized over more complex ones. This characteristic underscores the adaptive capability of these nonlinear model families, where solution complexity naturally scales with dataset complexity. As our previous work suggests [3,23,27], this bias emerges through an adaptive training process where output function complexity increases gradually.

6.2 Condensation: Mechanisms of near-optimistic performance

The ability of neural networks to achieve near-optimistic fitting performance can be theoretically understood through model rank and its relationship to the condensation phenomenon. We establish this connection first through a concrete example, then extend our analysis to broader settings.

Consider a simple two-neuron network with

$$f_\theta(x) = a_1 \tanh(w_1 x) + a_2 \tanh(w_2 x).$$

The model rank for this network is:

$$\text{rank}_{f_\theta}(\theta) = \dim(\text{span}\{\tanh(w_1 x), a_1 \tanh'(w_1 x)x, \tanh(w_2 x), a_2 \tanh'(w_2 x)x\}).$$

This formulation reveals a direct relationship between model rank and condensation:

- (i) When condensation occurs ($w_2 = \pm w_1$), the model rank satisfies $\text{rank}_{f_\theta}(\theta) \leq 2$.
- (ii) Without condensation ($w_2 \neq \pm w_1 \neq 0$ and $a_1, a_2 \neq 0$), the model rank is $\text{rank}_{f_\theta}(\theta) = 4$.

Thus, model rank provides a quantitative measure of condensation—lower rank indicates stronger condensation, as neurons effectively combine their functionality.

The model rank framework extends beyond neural networks, offering a unified theoretical perspective for understanding near-optimistic performance across model classes. While neurons are specific to neural networks, model rank provides a general measure of functional complexity applicable to deep models, matrix factorization models, and neural networks with various architectures and activation functions.

Recent research has shown that condensation naturally emerges during nonlinear training beyond the neural tangent kernel regime [9]. This phenomenon has been observed across multiple settings: two-layer ReLU networks on both synthetic and real datasets [12], three-layer architectures [26], and networks with diverse activation functions [27]. Training conditions that promote condensation – and thus lower model rank – include: large learning rates [1], dropout regularization [24], and small initialization (as demonstrated in Section 4.1).

Our theoretical framework suggests these training conditions facilitate near-optimistic performance by guiding models toward solutions with lower model rank. This connection

provides a mechanistic explanation for the effectiveness of certain training practices in achieving sample-efficient learning.

The path to near-optimistic performance thus involves the implicit regularization of model rank during training. This perspective unifies several key observations:

- (i) The effectiveness of specific training practices (initialization, learning rates, dropout).
- (ii) The emergence of condensation across different architectures.
- (iii) The achievement of sample-efficient learning in overparameterized models.

These connections suggest promising directions for theoretical investigation, particularly in understanding how different training regimes influence the model rank trajectory and, consequently, the sample complexity of learning.

7 Conclusions and discussion

In this work, we have developed an optimistic estimate framework for analyzing nonlinear regression, providing quantitative bounds on the sample complexity of overparameterized models. Through theoretical analysis and empirical validation, we have demonstrated the framework's utility in understanding the behavior of deep models, matrix factorization models and neural networks with both fully-connected and convolutional architectures.

Our analysis suggests that many nonlinear models can be recovered under heavy overparameterization. While these suggestions require further theoretical and empirical investigation, our framework offers novel perspectives into nonlinear regression:

- (1) **Characterization of Overparameterized Fitting:** We identify that models with non-constant optimistic sample sizes exhibit distinct fitting behaviors in the overparameterized regime. This provides a theoretical basis for distinguishing between different model classes based on their sample complexity characteristics.
- (2) **Parameter Efficiency Measures:** Building on Breiman's foundational question regarding the effective size of parameters of neural network [4], our framework provides a quantitative measure through $R_{f_\theta}(f)$ for target functions $f \in \mathcal{F}$. This metric offers a way to analyze parameter utilization across different model architectures.
- (3) **Model Inductive Bias:** Our analysis reveals that nonlinear models tend to favor solutions with lower optimistic sample sizes, providing a theoretical perspective on the implicit regularization observed in practice.
- (4) **Neural Network Architecture Analysis:** The framework offers insights into architectural choices: (i) free expressiveness in width, i.e. expressiveness of a NN can be improved through widening without increasing the optimistic sample size; (ii) costly expressiveness in connection, i.e. removing unnecessary connections between neurons substantially reduces the optimistic sample size.

These perspectives complement existing frameworks on nonlinear regression for analyzing the model's generalization ability. Several promising directions for future research emerge from this analysis:

- (i) understanding the relationship between optimistic estimates and global training dynamics, particularly how condensation emerges and evolves throughout the training process;
- (ii) extending our theoretical framework to more complex architectures and modern neural network designs, including multi-head attention mechanisms, transformers [19], and Kolmogorov-Arnold networks [11];
- (iii) developing principled approaches to design sample-efficient architectures and training techniques for specific tasks by leveraging our understanding of model rank and condensation.

Appendix A Estimate of the optimistic sample sizes

Definition A.1 (Model Rank). *Given any differentiable (in parameters) model f_θ , the model rank for any $\theta^* \in \mathbb{R}^M$ is defined as*

$$R_{f_\theta}(\theta^*) := \dim(\text{span}\{\partial_{\theta_i} f(\cdot; \theta^*)\}_{i=1}^M), \quad (\text{A.1})$$

where

$$\text{span}\{\phi_i(\cdot)\}_{i=1}^M = \left\{ \sum_{i=1}^M a_i \phi_i(\cdot) \mid a_1, \dots, a_M \in \mathbb{R} \right\},$$

and $\dim(\cdot)$ returns the dimension of a linear function space. Then the model rank for any function $f^* \in \mathcal{F}$ with model function space $\mathcal{F} := \{f(\cdot; \theta) \mid \theta \in \mathbb{R}^M\}$ is defined as

$$R_{f_\theta}(f^*) := \min_{\theta' \in \Theta_{f^*}} R_{f_\theta}(\theta'), \quad (\text{A.2})$$

where the target set

$$\Theta_{f^*} := \{\theta \mid f(\cdot; \theta) = f^*; \theta \in \mathbb{R}^M\}.$$

Given a differentiable model f_θ , the optimistic estimate is obtained by the following two steps: estimate $R_{f_\theta}(\theta)$ for $\theta \in \mathbb{R}^M$; and estimate $R_{f_\theta}(f)$ for $f \in \mathcal{F}$. The difficulty of optimistic estimate depends on the complexity of model architecture. In the following subsections, we present the intuitive steps that quickly derive the optimistic sample sizes for deep models, matrix factorization models and two-layer tanh-NNs with various architectures.

A.1 Deep model

$$f_\theta(x) = \sum_{j=1}^d \left(a_j^{[1]} \cdots a_j^{[L]} \right) x_j, \quad \theta = (a_1, \dots, a_d), \quad a_j = \left[a_j^{[1]}, \dots, a_j^{[L]} \right]^\top \in \mathbb{R}^L, \quad L \geq 2.$$

Proposition A.1. *Given the depth-2 model*

$$f_{\theta}(\mathbf{x}) = \sum_{j=1}^d (a_j^2 - b_j^2) x_j$$

for any k -sparse target function f^* (i.e. a function with exactly k nonzero coefficients in its expression), the optimistic sample size is

$$R_{f_{\theta}}(f^*) = k. \quad (\text{A.3})$$

Proof. Step 1. **Estimate $R_{f_{\theta}}(\theta)$.** For any parameter point θ , the tangent space is

$$\begin{aligned} \text{span} \left\{ \frac{\partial f_{\theta}(\mathbf{x})}{\partial a_j^{[l]}} \right\}_{j \in [d], l \in [L]} &= \text{span} \left\{ \left(\prod_{s \in [L] \setminus \{l\}} a_j^{[s]} \right) x_j \right\}_{j \in [d], l \in [L]} \\ &= \bigoplus_{j=1}^d \text{span} \left\{ \left(\prod_{s \in [L] \setminus \{l\}} a_j^{[s]} \right) x_j \right\}_{l=1}^L. \end{aligned}$$

We observe that, for any $j \in [d]$, if $\mathbf{a}_j$ has more than one zeros, then

$$\text{span} \left\{ \left(\prod_{s \in [L] \setminus \{l\}} a_j^{[s]} \right) x_j \right\}_{l=1}^L = \text{span}\{0(\cdot)\}.$$

Otherwise,

$$\text{span} \left\{ \left(\prod_{s \in [L] \setminus \{l\}} a_j^{[s]} \right) x_j \right\}_{l=1}^L = \text{span}\{x_j\}.$$

We define a function $g(\cdot)$ as $g(\mathbf{v}) = 0$ for vector $\mathbf{v}$ having more than one zeros, and $g(\mathbf{v}) = 1$ for vector $\mathbf{v}$ having no more than one zero. Then, we obtain

$$\begin{aligned} \dim \left(\text{span} \left\{ \left(\prod_{s \in [L] \setminus \{l\}} a_j^{[s]} \right) x_j \right\}_{l=1}^L \right) &= g(\mathbf{a}_j), \\ R_{f_{\theta}}(\theta) &= \dim \left\{ \bigoplus_{j=1}^d \text{span} \left\{ \left(\prod_{s \in [L] \setminus \{l\}} a_j^{[s]} \right) x_j \right\}_{l=1}^L \right\} = \sum_{j=1}^d g(\mathbf{a}_j). \end{aligned}$$

Step 2. **Estimate $R_{f_{\theta}}(f)$.**

Given any target function

$$f^* \in \mathcal{F} = \left\{ \sum_{j=1}^d \alpha_j x_j \mid \alpha_1, \dots, \alpha_d \in \mathbb{R} \right\},$$

we suppose it is k -sparse, i.e. it has k nonzero coefficients. Then, for any $\theta \in \Theta_{f^*}$, $\sum_{j=1}^d g(a_j) \geq k$. Specifically, $\sum_{j=1}^d g(a_j)$ attains k when $a_j = 0$ for all dimension j with zero coefficients. Therefore, we have

$$R_{f_\theta}(f^*) = k.$$

The proof is complete. $\square$

A.2 Matrix factorization model

$$f_\theta = AB, \quad \theta = (A, B), \quad A = [a_{ij}]_{i,j=1}^d \in \mathbb{R}^{d \times d}, \quad B = [b_{ij}]_{i,j=1}^d \in \mathbb{R}^{d \times d}.$$

Proposition A.2. For the matrix factorization model $f_\theta = AB$, the optimistic sample size for any $M^* \in \mathbb{R}^{d \times d}$ as

$$R_{f_\theta}(M^*) = 2r_{M^*}d - r_{M^*}^2, \quad (\text{A.4})$$

where $r_{M^*} = \text{rank}(M^*)$ is the matrix rank of M^* .

Proof. Step 1. Estimate $R_{f_\theta}(\theta)$.

For any parameter point θ , the tangent space is $\text{span}\{P^{ij}, Q^{ij}\}_{i,j=1}^d$, where

$$P^{ij} = \frac{\partial f_\theta}{\partial a_{ij}} = [\delta_{ki}b_{jl}]_{k,l=1}^d, \quad Q^{ij} = \frac{\partial f_\theta}{\partial b_{ij}} = [a_{ki}\delta_{jl}]_{k,l=1}^d.$$

Here δ_{ij} is the Kronecker delta. By vectorizing $P_{d \times d}^{ij}$ and $Q_{d \times d}^{ij}$, we get

$$\begin{aligned} \text{vec}(P^{ij}) &= [P_{11}^{ij}, \dots, P_{1d}^{ij}, P_{21}^{ij}, \dots, P_{2d}^{ij}, \dots, P_{d1}^{ij}, \dots, P_{dd}^{ij}]^\top \in \mathbb{R}^{d^2}, \\ \text{vec}(Q^{ij}) &= [Q_{11}^{ij}, \dots, Q_{1d}^{ij}, Q_{21}^{ij}, \dots, Q_{2d}^{ij}, \dots, Q_{d1}^{ij}, \dots, Q_{dd}^{ij}]^\top \in \mathbb{R}^{d^2}. \end{aligned}$$

Then, we have

$$R_{f_\theta}(\theta) = \dim(\text{span}\{P^{ij}, Q^{ij}\}_{i,j=1}^d) = \dim(\text{span}\{\text{vec}(P^{ij}), \text{vec}(Q^{ij})\}_{i,j=1}^d) = \text{rank}(\Gamma),$$

where

$$\Gamma = [\text{vec}(Q^{11}), \dots, \text{vec}(Q^{dd}), \text{vec}(P^{11}), \dots, \text{vec}(P^{dd})]^\top$$

is a $2d^2 \times d^2$ matrix. Using the Kronecker product $\otimes$ of matrices, we obtain

$$\Gamma = \begin{bmatrix} B & & & \\ & B & & \\ & & \ddots & \\ & & & B \\ a_{11}I & a_{21}I & \cdots & a_{d1}I \\ a_{12}I & a_{22}I & \cdots & a_{d2}I \\ \vdots & \vdots & \ddots & \vdots \\ a_{1d}I & a_{2d}I & \cdots & a_{dd}I \end{bmatrix} = \begin{bmatrix} I \otimes B \\ A^\top \otimes I \end{bmatrix}.$$

Then,

$$\text{rank}(\Gamma) = \text{rank}(\Gamma^\top \Gamma) = \text{rank}(I \otimes B^\top B + AA^\top \otimes I) = \text{rank}(B^\top B \oplus AA^\top),$$

where $\oplus$ indicates the Kronecker sum. Let $r_A := \text{rank}(A)$, $r_B := \text{rank}(B)$. Because the eigenvalues of $B^\top B \oplus AA^\top$ are the sums of all possible pairs of eigenvalues of $B^\top B$ and $AA^\top$, there are $(d - r_A)(d - r_B)$ zero eigenvalues. Therefore, we have

$$R_{f_\theta}(\theta) = \text{rank}(B^\top B \oplus AA^\top) = d^2 - (d - r_A)(d - r_B). \quad (\text{A.5})$$

Step 2. Estimate $R_{f_\theta}(f)$.

Given any target matrix $f^* \in \mathbb{R}^{d \times d}$, let $r = \text{rank}(f^*)$. Because

$$\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\},$$

any factorization $f^* = A^*B^*$ satisfies $\text{rank}(A^*) \geq r$ and $\text{rank}(B^*) \geq r$. By Eq. (A.5),

$$R_{f_\theta}(\theta^*) \geq d^2 - (d - r)^2 = 2rd - r^2.$$

This lower bound can be obtained using singular value decomposition $f^* = U\Sigma V^\top$, and let $\theta^* = (A^*, B^*)$ with $A^* = U\Sigma^{1/2}$ and $B^* = \Sigma^{1/2}V^\top$. Then,

$$R_{f_\theta}(f^*) = R_{f_\theta}(\theta^*) = 2rd - r^2. \quad (\text{A.6})$$

The proof is complete. $\square$

A.3 Two-layer fully-connected NN

$$f_\theta(x) = \sum_{i=1}^m a_i \sigma(w_i^\top x), \quad x = [x_1, \dots, x_d]^\top \in \mathbb{R}^d, \quad \theta = (a_i \in \mathbb{R}, w_i \in \mathbb{R}^d)_{i=1}^m.$$

Proposition A.3. *Given the two-layer fully-connected neural network*

$$f_\theta(x) = \sum_{i=1}^m a_i \sigma(w_i^\top x)$$

with σ satisfying Assumption 5.1. For any f^* expressible by the width- m NN, denote its intrinsic width as $0 \leq k(f^*) \leq m$, the optimistic sample size of f^* is

$$R_{\text{NN}_m}(f^*) = k(f^*)(d + 1). \quad (\text{A.7})$$

Proof. **Step 1. Estimate $R_{\text{NN}_m}(\theta)$.**

For any parameter point θ , the tangent space is

$$\begin{aligned} & \text{span} \left\{ \frac{\partial f_\theta(x)}{\partial a_i}, \frac{\partial f_\theta(x)}{\partial w_{i1}}, \dots, \frac{\partial f_\theta(x)}{\partial w_{id}} \right\}_{i=1}^m \\ &= \text{span} \{ \sigma(w_i^\top x), a_i \sigma'(w_i^\top x) x_1, \dots, a_i \sigma'(w_i^\top x) x_d \}_{i=1}^m \\ &= \text{span} \{ \sigma(w_i^\top x) \}_{i=1}^m \oplus \text{span} \{ a_i \sigma'(w_i^\top x) x_1 \}_{i=1}^m \oplus \dots \oplus \text{span} \{ a_i \sigma'(w_i^\top x) x_d \}_{i=1}^m. \end{aligned}$$

Then

$$\begin{aligned} & \dim \left(\text{span} \left\{ \frac{\partial f_{\theta}(\mathbf{x})}{\partial a_i}, \frac{\partial f_{\theta}(\mathbf{x})}{\partial w_{i1}}, \dots, \frac{\partial f_{\theta}(\mathbf{x})}{\partial w_{id}} \right\}_{i=1}^m \right) \\ &= \dim \left(\text{span} \{ \sigma(\mathbf{w}_i^{\top} \mathbf{x}) \}_{i=1}^m \right) + \dim \left(\text{span} \{ a_i \sigma'(\mathbf{w}_i^{\top} \mathbf{x}) x_1 \}_{i=1}^m \right) d. \end{aligned}$$

Let

$$m_w = \dim \left(\text{span} \{ \sigma(\mathbf{w}_i^{\top} \mathbf{x}) \}_{i=1}^m \right), \quad m_a = \dim \left(\text{span} \{ a_i \sigma'(\mathbf{w}_i^{\top} \mathbf{x}) x_1 \}_{i=1}^m \right),$$

we obtain

$$R_{\text{NN}_m}(\theta) = m_w + m_a d. \quad (\text{A.8})$$

Step 2. Estimate $R_{\text{NN}_m}(f)$.

Given any $f^* \in \mathcal{F}_m^{\text{NN}}$, we suppose $f^* \in \mathcal{F}_k^{\text{NN}} / \mathcal{F}_{k-1}^{\text{NN}}$, i.e., the intrinsic width $k(f^*) = k$. Then, for any $\theta \in \Theta_{f^*}$, we have $m_w \geq k$ and $m_a \geq k$. Therefore, $R_{\text{NN}_m}(\theta) \geq k(d+1)$. R_{NN_m} attains $k(d+1)$ for a $\theta^* \in \Theta_{f^*}$ constructed as follows. Because $f^* \in \mathcal{F}_k^{\text{NN}} / \mathcal{F}_{k-1}^{\text{NN}}$, it can be expressed by a width- k NN as $\sum_{i=1}^k a'_i \sigma(\mathbf{w}'_i{}^{\top} \mathbf{x})$ satisfying (i) $a'_i \neq 0$ and $\mathbf{w}'_i \neq \mathbf{0}$ for all $i \in [k]$ and (ii) $\mathbf{w}'_i \neq \mathbf{w}'_j$ for any pair $i, j \in [k]$ and $i \neq j$. Then

$$\theta^* = (a'_1, \mathbf{w}'_1, \dots, a'_k, \mathbf{w}'_k, 0, \mathbf{0}, \dots, 0, \mathbf{0}) \quad (\text{A.9})$$

satisfies $R_{\text{NN}_m}(\theta^*) = k(d+1)$. Therefore, we obtain

$$R_{\text{NN}_m}(f^*) = k(f^*)(d+1). \quad (\text{A.10})$$

The proof is complete. $\square$

A.4 Two-layer convolutional NN

$$f_{\theta}(\mathbf{x}) = \sum_{i=1}^{m_C} \sum_{j=1}^{d+1-s} a_{ij} \sigma \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{i;\alpha} \right)$$

with $\mathbf{x} = [x_1, \dots, x_d]^{\top} \in \mathbb{R}^d$, $\theta = (\mathbf{a}_i \in \mathbb{R}^{d+1-s}, \mathbf{K}_i \in \mathbb{R}^s)_{i=1}^{m_C}$.

Proposition A.4. *Given the two-layer CNN with weight-sharing*

$$f_{\theta}(\mathbf{x}) = \sum_{i=1}^{m_C} \sum_{j=1}^{d+1-s} a_{ij} \sigma \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{i;\alpha} \right)$$

with σ satisfying Assumption 5.1. For any f^* expressible by the width- m CNN, denote its intrinsic width as $0 \leq k_C(f^*) \leq m$, the optimistic sample size of f^* is

$$R_{\text{CNN}_{m_C}}(f^*) = k_C(f^*)(d+1). \quad (\text{A.11})$$

Proof. Step 1. **Estimate** $R_{\text{CNN}_{m_C}}(\theta)$.

For any parameter point θ , the tangent space is

$$\begin{aligned}
 & \text{span} \left\{ \frac{\partial f_{\theta}(\mathbf{x})}{\partial a_{i1}}, \dots, \frac{\partial f_{\theta}(\mathbf{x})}{\partial a_{i(d+1-s)}}, \frac{\partial f_{\theta}(\mathbf{x})}{\partial K_{i1}}, \dots, \frac{\partial f_{\theta}(\mathbf{x})}{\partial K_{is}} \right\}_{i=1}^{m_C} \\
 &= \text{span} \left\{ \sigma \left(\sum_{\alpha=1}^s x_{1+s-\alpha} K_{i;\alpha} \right), \dots, \sigma \left(\sum_{\alpha=1}^s x_{d+1-\alpha} K_{i;\alpha} \right), \right. \\
 & \quad \left. \sum_{j=1}^{d+1-s} a_{ij} \sigma' \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{i;\alpha} \right) x_{j+s-1}, \dots, \sum_{j=1}^{d+1-s} a_{ij} \sigma' \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{i;\alpha} \right) x_j \right\}_{i=1}^{m_C} \\
 &= \text{span} \left\{ \sigma \left(\sum_{\alpha=1}^s x_{1+s-\alpha} K_{i;\alpha} \right) \right\}_{i=1}^{m_C} \oplus \dots \oplus \text{span} \left\{ \sigma \left(\sum_{\alpha=1}^s x_{d+1-\alpha} K_{i;\alpha} \right) \right\}_{i=1}^{m_C} \\
 & \quad \oplus \text{span} \left\{ \sum_{j=1}^{d+1-s} a_{ij} \sigma' \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{i;\alpha} \right) x_{j+s-1} \right\}_{i=1}^{m_C} \\
 & \quad \oplus \dots \oplus \text{span} \left\{ \sum_{j=1}^{d+1-s} a_{ij} \sigma' \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{i;\alpha} \right) x_j \right\}_{i=1}^{m_C}.
 \end{aligned}$$

Then

$$\begin{aligned}
 & \dim \left(\text{span} \left\{ \frac{\partial f_{\theta}(\mathbf{x})}{\partial a_{i1}}, \dots, \frac{\partial f_{\theta}(\mathbf{x})}{\partial a_{i(d+1-s)}}, \frac{\partial f_{\theta}(\mathbf{x})}{\partial K_{i1}}, \dots, \frac{\partial f_{\theta}(\mathbf{x})}{\partial K_{is}} \right\}_{i=1}^{m_C} \right) \\
 &= \dim \left(\text{span} \left\{ \sigma \left(\sum_{\alpha=1}^s x_{1+s-\alpha} K_{i;\alpha} \right) \right\}_{i=1}^{m_C} \right) (d+1-s) \\
 & \quad + \dim \left(\text{span} \left\{ \sum_{j=1}^{d+1-s} a_{ij} \sigma' \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{i;\alpha} \right) x_j \right\}_{i=1}^{m_C} \right) s.
 \end{aligned}$$

Let

$$\begin{aligned}
 m_K^C &= \dim \left(\text{span} \left\{ \sigma \left(\sum_{\alpha=1}^s x_{1+s-\alpha} K_{i;\alpha} \right) \right\}_{i=1}^{m_C} \right), \\
 m_a^C &= \dim \left(\text{span} \left\{ \sum_{j=1}^{d+1-s} a_{ij} \sigma' \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{i;\alpha} \right) x_j \right\}_{i=1}^{m_C} \right),
 \end{aligned}$$

we obtain

$$R_{\text{CNN}_{m_C}}(\theta) = m_K^C(d+1-s) + m_a^C s. \quad (\text{A.12})$$

Step 2. **Estimate** $R_{\text{CNN}_{m_C}}(f)$.

Given any $f^* \in \mathcal{F}_{m_C}^{\text{CNN}}$, we suppose $f^* \in \mathcal{F}_k^{\text{CNN}} / \mathcal{F}_{k-1}^{\text{CNN}}$, i.e. the intrinsic number of kernels $k_C(f^*) = k$. Then, for any $\theta \in \Theta_{f^*}$, we have $m_K^C \geq k$ and $m_a^C \geq k$. Therefore, $R_{\text{CNN}_{m_C}}(\theta) \geq k(d+1)$. $R_{\text{CNN}_{m_C}}$ attains $k(d+1)$ for a $\theta^* \in \Theta_{f^*}$ constructed as follows. Because $f^* \in \mathcal{F}_k^{\text{CNN}} / \mathcal{F}_{k-1}^{\text{CNN}}$, it can be expressed by a width- k CNN as

$$\sum_{i=1}^k \sum_{j=1}^{d+1-s} a'_{ij} \sigma \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K'_{i;\alpha} \right),$$

satisfying (i) $a'_i \neq \mathbf{0}$ and $K'_i \neq \mathbf{0}$ for all $i \in [k]$, and (ii) $K'_i \neq K'_j$ for any pair $i, j \in [k]$ and $i \neq j$. Then

$$\theta^* = (a'_1, K'_1, \dots, a'_k, K'_k, 0, 0, \dots, 0, 0) \quad (\text{A.13})$$

satisfies $R_{\text{CNN}_{m_C}}(\theta^*) = k(d+1)$. Therefore, we obtain

$$R_{\text{CNN}_{m_C}}(f^*) = k_C(f^*)(d+1). \quad (\text{A.14})$$

The proof is complete. $\square$

Following a similar procedure, we can also obtain the optimistic sample size for two-layer CNNs with 2-d convolution. Suppose the input dimension is $d \times d$ and the kernel size is $s \times s$ with stride 1, the optimistic sample size

$$R_{\text{CNN}_{m_C}}(f^*) = k_C(f^*)(s^2 + (d+1-s)^2). \quad (\text{A.15})$$

A.5 Two-layer no-sharing convolutional NN

$$f_{\theta}(x) = \sum_{i=1}^{m_C} \sum_{j=1}^{d+1-s} a_{ij} \sigma \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{ij;\alpha} \right)$$

with $x = [x_1, \dots, x_d]^{\top} \in \mathbb{R}^d$, $\theta = (a_i \in \mathbb{R}^{d+1-s}, K_{ij} \in \mathbb{R}^s)_{i=1}^{m_C}$.

Proposition A.5. *Given the two-layer CNN without weight-sharing*

$$f_{\theta}(x) = \sum_{i=1}^{m_C} \sum_{j=1}^{d+1-s} a_{ij} \sigma \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{ij;\alpha} \right)$$

with σ satisfying Assumption 5.1. For any f^* expressible by the width- m CNN without weight sharing, denote its intrinsic width as $0 \leq k_C(f^*) \leq m$, the optimistic sample size of f^* is

$$R_{\text{CNN}_{m_C}^{\text{NS}}}(f^*) = k_C(f^*)(d+1-s)(s+1). \quad (\text{A.16})$$

Proof. Step 1. **Estimate** $R_{\text{CNN}_{m_C}^{\text{NS}}}(\theta)$.

For any parameter point θ , the tangent space is

$$\begin{aligned}
 & \text{span} \left\{ \frac{\partial f_{\theta}(x)}{\partial a_{ij}}, \frac{\partial f_{\theta}(x)}{\partial K_{ij;1}}, \dots, \frac{\partial f_{\theta}(x)}{\partial K_{ij;s}} \right\}_{i \in [m_C], j \in [d+1-s]} \\
 &= \text{span} \left\{ \sigma \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{ij;\alpha} \right), a_{ij} \sigma' \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{ij;\alpha} \right) x_{j+s-1}, \dots, \right. \\
 & \quad \left. a_{ij} \sigma' \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{ij;\alpha} \right) x_j \right\}_{i \in [m_C], j \in [d+1-s]} \\
 &= \text{span} \left\{ \sigma \left(\sum_{\alpha=1}^s x_{1+s-\alpha} K_{ij;\alpha} \right) \right\}_{i \in [m_C], j \in [d+1-s]} \\
 & \quad \oplus \text{span} \left\{ a_{ij} \sigma' \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{ij;\alpha} \right) x_{j+s-1} \right\}_{i \in [m_C], j \in [d+1-s]} \\
 & \quad \oplus \dots \oplus \text{span} \left\{ a_{ij} \sigma' \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{ij;\alpha} \right) x_j \right\}_{i \in [m_C], j \in [d+1-s]}.
 \end{aligned}$$

Let

$$\begin{aligned}
 m_K &= \dim \left(\text{span} \left\{ \sigma \left(\sum_{\alpha=1}^s x_{1+s-\alpha} K_{ij;\alpha} \right) \right\}_{i \in [m_C], j \in [d+1-s]} \right), \\
 m_a &= \dim \left(\text{span} \left\{ a_{ij} \sigma' \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K_{ij;\alpha} \right) x_j \right\}_{i \in [m_C], j \in [d+1-s]} \right),
 \end{aligned}$$

we obtain

$$\begin{aligned}
 R_{\text{CNN}_{m_C}^{\text{NS}}}(\theta) &= \dim \left(\text{span} \left\{ \frac{\partial f_{\theta}(x)}{\partial a_{ij}}, \frac{\partial f_{\theta}(x)}{\partial K_{ij;1}}, \dots, \frac{\partial f_{\theta}(x)}{\partial K_{ij;s}} \right\}_{i \in [m_C], j \in [d+1-s]} \right) \\
 &= m_w + m_a s.
 \end{aligned} \tag{A.17}$$

Step 2. **Estimate** $R_{\text{CNN}_{m_C}^{\text{NS}}}(f)$ **for a subset of** $\mathcal{F}_{m_C}^{\text{CNN-NS}}$.

Given any $f^* \in \mathcal{F}_{m_C}^{\text{CNN-NS}}$, we suppose $f^* \in \mathcal{F}_k^{\text{CNN-NS}} / \mathcal{F}_{k-1}^{\text{CNN-NS}}$, i.e. the intrinsic number of kernels $k_C(f^*) = k$. Then f^* can be expressed as

$$\sum_{i=1}^k \sum_{j=1}^{d+1-s} a'_{ij} \sigma \left(\sum_{\alpha=1}^s x_{j+s-\alpha} K'_{ij;\alpha} \right).$$

If f^* further satisfies

- (i) $a'_{ij} \neq 0$ and $K'_{ij} \neq \mathbf{0}$ for all $i \in [k], j \in [d+1-s]$.
- (ii)

$$\sigma\left(\sum_{\alpha=1}^s x_{j+s-\alpha} K'_{ij;\alpha}\right) \neq \sigma\left(\sum_{\alpha=1}^s x_{h+s-\alpha} K'_{lh;\alpha}\right)$$

for all pairs $(i, j) \neq (l, h)$, then $m_K \geq k(d+1-s)$ and $m_a \geq k(d+1-s)$.

Therefore, $R_{\text{CNN}_{m_C}^{\text{NS}}}(\theta) \geq k(d+1-s)(s+1)$. $R_{\text{CNN}_{m_C}^{\text{NS}}}$ attains $k(d+1)$ for

$$\theta^* = (a'_{11}, K'_{11}, \dots, a'_{k(d+1-s)}, K'_{k(d+1-s)}, 0, \mathbf{0}, \dots, 0, \mathbf{0}). \quad (\text{A.18})$$

Therefore, we obtain

$$R_{\text{CNN}_{m_C}^{\text{NS}}}(f^*) = k(d+1-s)(s+1). \quad (\text{A.19})$$

The proof is complete. $\square$

Following a similar procedure, we can also obtain the optimistic sample size for two-layer no-sharing CNNs with 2-d convolution. Suppose the input dimension is $d \times d$ and the kernel size is $s \times s$ with stride 1, the optimistic sample size

$$R_{\text{CNN}_{m_C}^{\text{NS}}}(f^*) = k(d+1-s)^2(s^2+1). \quad (\text{A.20})$$

Appendix B Details of experiments

In Fig. 3.1, model parameters are initialized with a normal distribution with mean 0 and variance 10^{-8} , and trained with full-batch GD with a learning rate of 0.01. The training ends when the training error is less than 10^{-9} .

In Fig. 4.1, model parameters are initialized with a normal distribution with mean 0 and variance 10^{-8} , and trained with full-batch GD with a learning rate of 0.05. The training ends when the training error is less than 10^{-9} . The target matrices we use are as follows:

$$\begin{aligned} M_1^* &= \begin{bmatrix} 1 & 0.3 & 0.7 & -0.4 \\ 2 & 0.6 & 1.4 & -0.8 \\ 4 & 1.2 & 2.8 & -1.6 \\ 7 & 2.1 & 4.9 & -2.8 \end{bmatrix}, & M_2^* &= \begin{bmatrix} 4 & 0.6 & 1.8 & 0.8 \\ 6 & 0.9 & 2.7 & 1.2 \\ 8 & 1.2 & 3.6 & 1.6 \\ 18 & 2.7 & 8.1 & 3.6 \end{bmatrix}, \\ M_3^* &= \begin{bmatrix} -1.8 & 2.4 & 7.7 & -5.3 \\ 0.4 & 1.8 & 5.4 & -3.6 \\ 3.2 & 1.8 & 4.8 & -3 \\ 6.6 & 2.4 & 5.9 & -3.5 \end{bmatrix}, & M_4^* &= \begin{bmatrix} 7.6 & 3.3 & 19.8 & -7.3 \\ 7.6 & 2.1 & 10.7 & -2.4 \\ 8.8 & 1.8 & 7.6 & -0.2 \\ 19.2 & 3.6 & 14.1 & 0.9 \end{bmatrix}, \\ M_5^* &= \begin{bmatrix} -1.8 & 2.4 & 7.7 & -5.3 \\ 0.4 & 1.8 & 5.4 & -3.6 \\ 3.2 & 1.8 & 4.8 & -3 \\ 6.6 & 2.4 & 5.9 & 3.5 \end{bmatrix}, & M_6^* &= \begin{bmatrix} 8.5 & 9.3 & 22.5 & -6.1 \\ 8.2 & 6.1 & 12.5 & -1.6 \\ 11.5 & 19.8 & 15.7 & 3.4 \\ 20.4 & 11.6 & 17.7 & 2.5 \end{bmatrix}, \end{aligned}$$

$$M_7^* = \begin{bmatrix} 3.6 & -1.2 & 8.1 & -3.5 \\ 8.1 & -3.5 & 3.6 & -1.2 \\ 9.1 & -1.7 & 11.4 & -0.6 \\ 11.4 & -0.6 & 9.1 & -1.7 \end{bmatrix}, \quad M_8^* = \begin{bmatrix} 12.1 & 17.3 & 24.1 & -4.9 \\ 16.3 & 24.1 & 16.1 & 1.1 \\ 14.2 & 25.8 & 16.9 & 4.3 \\ 22.2 & 15.6 & 18.5 & 3.1 \end{bmatrix}.$$

In Fig. 4.2(b), We use M_1^* above as the target matrix. For all experiments, the network is initialized by a normal distribution with mean 0 and variance σ^2 , and trained with full-batch GD with a learning rate 0.05.

In Fig. 5.1, we use fully-connected NNs or CNNs with bias term b , e.g.

$$f_{\theta}(x) = \sum_{i=1}^m a_i \tanh(w_i^{\top} x + b_i),$$

to fit the target function. Therefore, a m -kernel CNN has $7m$ parameters, and the corresponding width- $3m$ fully-connected NN has $21m$ parameters. Fully-connected NNs and CNNs with $m = 1, 3, 10, 34$, or 100 are used for fitting. Network parameters are initialized by a normal distribution with mean 0 and variance 10^{-20} , and trained by full-batch gradient descent. The learning rate for the experiments in each setup is fine-tuned from 0.05 to 0.5 for a better generalization performance. For the training dataset and the test dataset, we construct the input data through the standard normal distribution and obtain the output values from the target function. The size of the training dataset varies whereas the size of the test dataset is fixed to 1000.

In Fig. B.1, we visualize the sequential sampling pattern where the number of samples (n) increases incrementally from 1 to 16 used in Fig. 6.1. The numerical value in each cell indicates the specific iteration at which that location was sampled, with values ranging from 1 to n . The color gradient provides an intuitive representation of the sampling sequence: red hues indicate earlier sampling (lower n values), while blue hues represent later sampling (higher n values). This visualization employs a heat map color scheme where warmer colors (redder) mean earlier sampling points and cooler colors (bluer) mean later sampling points. The figure effectively demonstrates the spatial and temporal distribution of the sampling process across all 16 iterations.



Figure B.1: Visualization of the sequential sampling pattern. The color gradient transitions from red (early samples) to blue (later samples), with numerical values indicating the specific sampling iteration.

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References

- [1] M. Andriushchenko, A. Varre, L. Pillaud-Vivien, and N. Flammarion, SGD with large step sizes learns sparse features, *arXiv:2210.05337*, 2022.
- [2] S. Arora, N. Cohen, W. Hu, and Y. Luo, Implicit regularization in deep matrix factorization, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 32:7381–7392, 2019.
- [3] Z. Bai, T. Luo, Z.-Q. J. Xu, and Y. Zhang, Embedding principle in depth for the loss landscape analysis of deep neural networks, *arXiv:2205.13283*, 2022.
- [4] L. Breiman, Reflections after refereeing papers for NIPS, in: *The Mathematics of Generalization*, Westview Press, 11–15, 1995.
- [5] T. Brown et al., Language models are few-shot learners, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 33:1877–1901, 2020.
- [6] E. J. Candès and M. B. Wakin, An introduction to compressive sampling, *IEEE Signal Process. Mag.*, 25(2):21–30, 2008.
- [7] Y. Cooper, Global minima of overparameterized neural networks, *SIAM J. Math. Data Sci.*, 3(2):676–691, 2021.
- [8] S. Gunasekar, B. E. Woodworth, S. Bhojanapalli, B. Neyshabur, and N. Srebro, Implicit regularization in matrix factorization, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 30:6152–6160, 2017.
- [9] A. Jacot, F. Gabriel, and C. Hongler, Neural tangent kernel: Convergence and generalization in neural networks, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 31:8571–8580, 2018.
- [10] J. Jumper et al., Highly accurate protein structure prediction with AlphaFold, *Nature*, 596(7873):583–589, 2021.
- [11] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark, KAN: Kolmogorov-Arnold networks, *arXiv:2404.19756*, 2024.
- [12] T. Luo, Z.-Q. J. Xu, Z. Ma, and Y. Zhang, Phase diagram for two-layer ReLU neural networks at infinite-width limit, *J. Mach. Learn. Res.*, 22(1):71, 2021.
- [13] D. MacKay, Bayesian model comparison and backprop nets, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 4:839–846, 1991.
- [14] W. J. Maddox, G. Benton, and A. G. Wilson, Rethinking parameter counting in deep models: Effective dimensionality revisited, *arXiv:2003.02139*, 2020.
- [15] OpenAI et al., GPT-4 Technical Report, *arXiv:2303.08774*, 2023.
- [16] S. Shalev-Shwartz and S. Ben-David, *Understanding Machine Learning: From Theory to Algorithms*, Cambridge University Press, 2014.
- [17] D. Silver et al., Mastering the game of Go with deep neural networks and tree search, *Nature*, 529(7587):484–489, 2016.
- [18] N. Srebro, *Learning with Matrix Factorizations*, PhD Thesis, Massachusetts Institute of Technology, 2004.

- [19] A. Vaswani, Attention is all you need, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 30:5999–6009, 2017.
- [20] D. H. Wolpert and W. G. Macready, No free lunch theorems for optimization, *IEEE Trans. Evol. Comput.*, 1(1):67–82, 1997.
- [21] B. Woodworth, S. Gunasekar, J. D. Lee, E. Moroshko, P. Savarese, I. Golan, D. Soudry, and N. Srebro, Kernel and rich regimes in overparametrized models, in: *Conference on Learning Theory*, PMLR, 3635–3673, 2020.
- [22] C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals, Understanding deep learning requires rethinking generalization, in: *Proceeding of 5th International Conference on Learning Representations*, Curran Associates, Inc., Vol.4, 3001–3015, 2017.
- [23] Y. Zhang, Z. Zhang, T. Luo, and Z. J. Xu, Embedding principle of loss landscape of deep neural networks, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 34:14848–14859, 2021.
- [24] Z. Zhang and Z.-Q. J. Xu, Implicit regularization of dropout, *IEEE Trans. Pattern Anal. Mach. Intell.*, 46(6):4206–4217, 2024.
- [25] K. Zhong, Z. Song, P. Jain, P. L. Bartlett, and I. S. Dhillon, Recovery guarantees for one-hidden-layer neural networks, in: *Proceedings of the 34th International Conference on Machine Learning*, Vol. 70, JMLR, 4140–4149, 2017.
- [26] H. Zhou, Q. Zhou, Z. Jin, T. Luo, Y. Zhang, and Z.-Q. J. Xu, Empirical phase diagram for three-layer neural networks with infinite width, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 35:26021–26033, 2022.
- [27] H. Zhou, Q. Zhou, T. Luo, Y. Zhang, and Z.-Q. J. Xu, Towards understanding the condensation of neural networks at initial training, in: *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 35:2184–2196, 2022.
- [28] E. R. Ziegel, The elements of statistical learning, *Technometrics*, 45(3):267–268, 2003.